Local Transport Automation System

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Motivation

- Most contemporary systems work in digital world.
- Digitizing local transport has multiple advantages from using digital money for ticket purchases to saving ones travel logs etc.
- Highly helpful when one traverses in a new city or an unknown route.

(a)

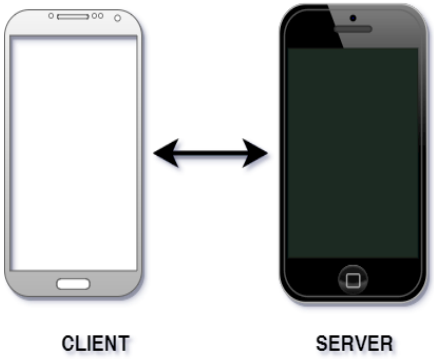
(b)

(c)

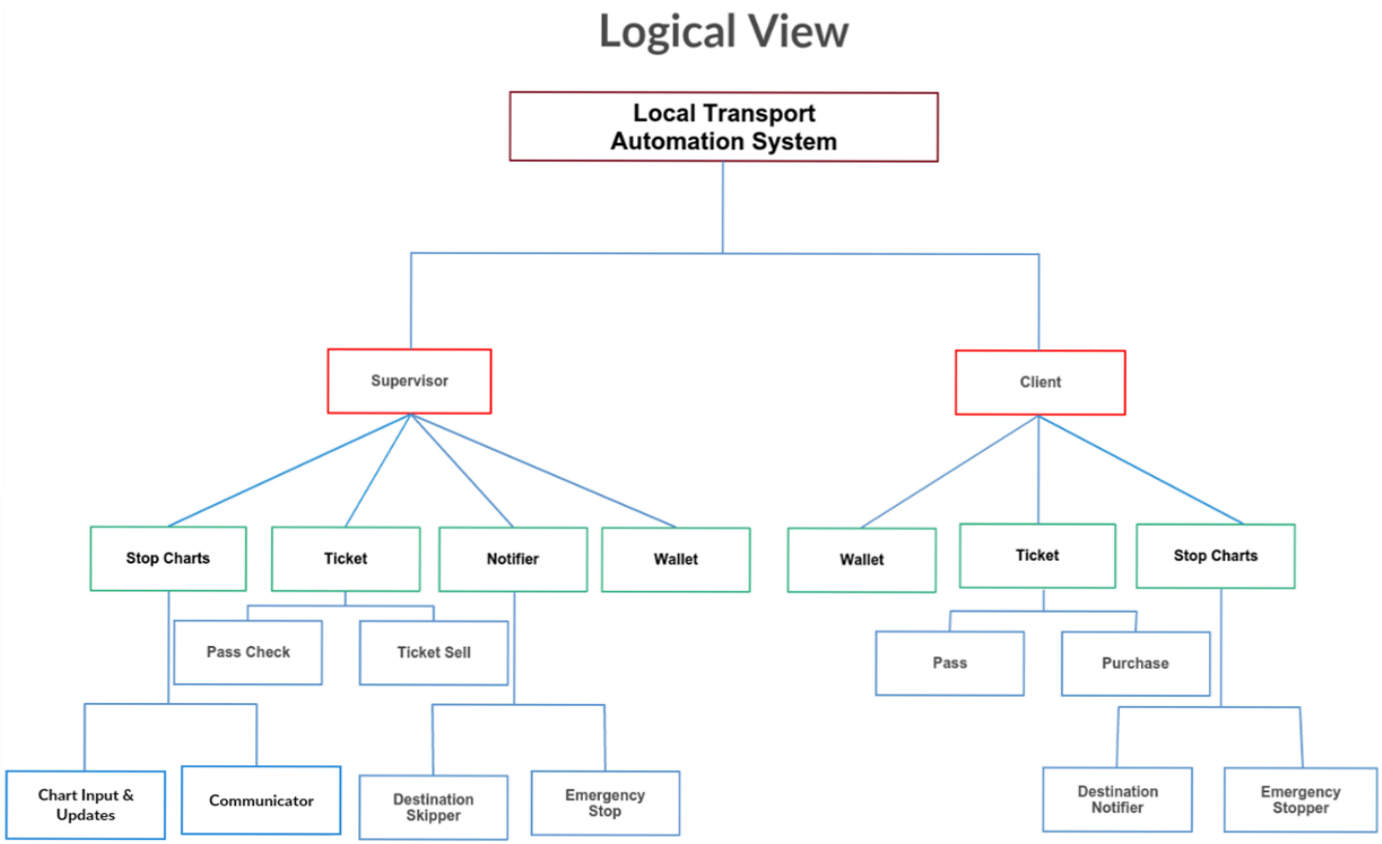
Figure 1: Digitization can abstract out Aspects

Problem Statement : Formulation

- Devise a system to handle multiple difficulties faced during local travel.
- Two android applications one for the bus supervisor and the other for the client are proposed with functionalities to buy/sell tickets, transfer bus-stop information etc.



Problem Statement : Functionality



Problem Statement : Functionality : Client

- **Wallet System** :
  - Provision for using digital money for buying tickets.
- **Digital Ticketing** :
  - The proposed project caters for digital ticketing by the 'Ticket' module under the client in the logical view.
- **Bus stop information** :
  - The bus-stops' information along the route of travel is provided by the 'Stop Charts' module.

Problem Statement : Functionality : Server

- **Wallet System** :
  - This wallet system is a parallel 'Wallet' module on the supervisor side too receive money for the client.
- **Digital ticket seller** :
  - This is again a parallel to 'Ticket' buyer on the client. This module sells tickets in return for money. This is also responsible for the pass verification.
- **Notifications to supervisor** :
  -'Destination Skipper' and the 'Emergency Stop' modules.

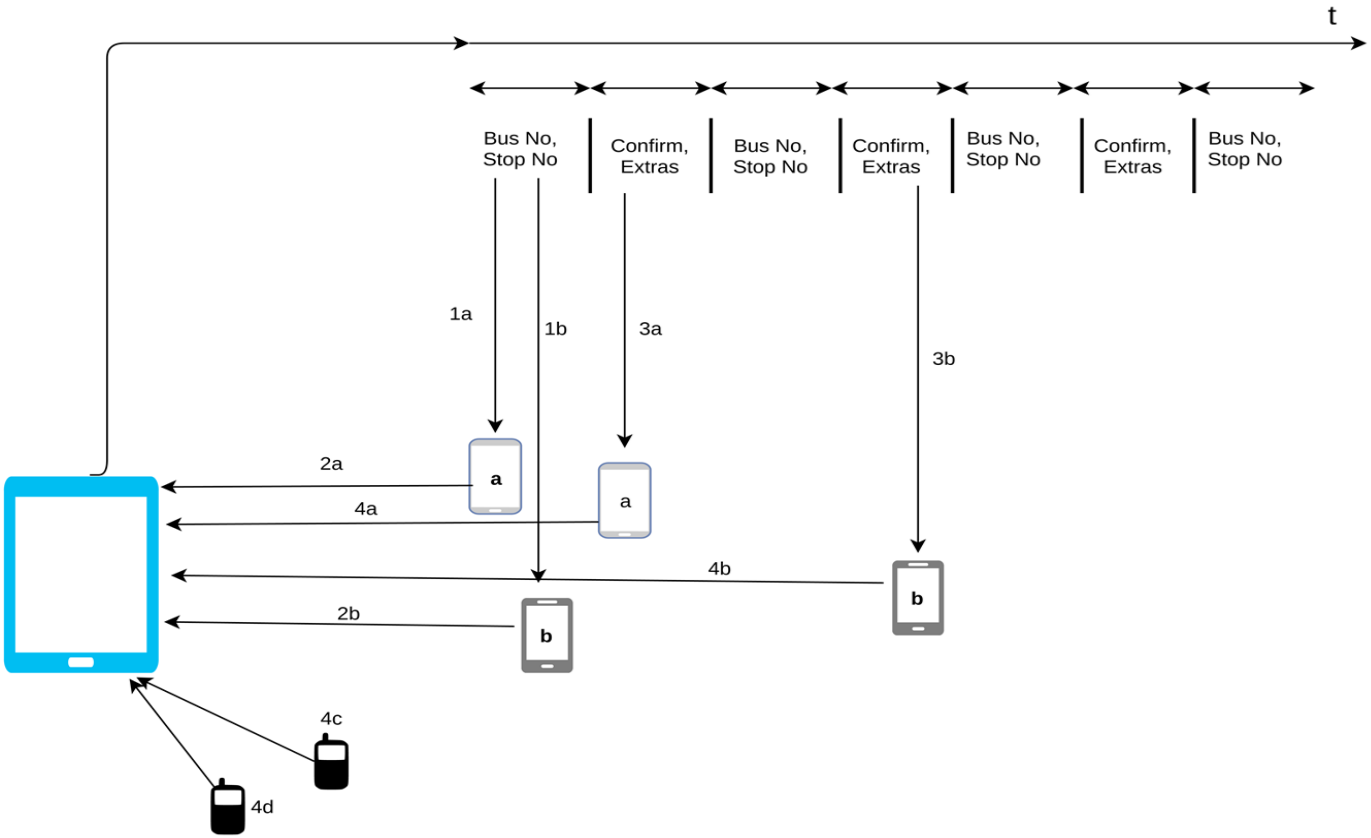
Problem Statement : Requirements : Solution

- **Bluetooth** :
  PROS
  - Low Power Consumption
  - Low Cost
  CONS
  - Low Bandwidth
  - Low Security
- Properly allocate time for multiple communication processes
- Secure the communication.
- ENCODE Messages in Device Name and broadcast. Continuosly DECODE and reply to incoming messages.

Problem Statement : Requirements

- **Server** :
  - Can receive Multiple requests.
  - Can send msg to limited mobiles at a time
- **Client** :
  - Can receive Multiple requests.
  - Requires to send one msg at a time.
- **Requirements** :
  - Handle multiple users.
  - Send multiple information..current state and confirmation messages
  - Internet limited to Server
  - Cost Efficient.

Architecture



Architecture : 4 Cycle Communication

- **Step 1** :
  - Basic information about the bus and current stop received by client
- **Step 2** :
  - Source and destination station sent - Emergency stop information can be sent
- **Step 3** :
  - Ticket confirmed
- **Step 4** :
  - Transaction confirmed by client

Architecture :

- **Detection** :
  - Both devices are continuosly trying to detect relevant messages.
- **Transmission** :
  - Client trasmits information right away after processing. - Server transmits after processing at specific time intervals.

Quality Attributes

- **Security** :
  - AES 128 bit encoding + Rijndael cipher
- **Interoperability** :
  - Communication system can be integrated into other systems low bandwidth scenarios with minor modifications.
- **Scalability** :
  - 4 step communication ensures smooth working at high traffic levels
- **Usability** :
  - Simple to use UI and intuitive concepts

Quality Attributes : Developer

- **Testability**
- **Maintainability**
- **Reusability**
- **Portability**

Demo

- **DEMO**
- **Questions ??**
- **Thank you**